

EGH404 ASSESSMENT 3: DATA ANALYSIS REPORT

EGH404: Research in Engineering Practice



SEMESTER 2, 2024

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EGH 404 Assessment 3B – Data Analysis

This is the template for Assessment 3B Data Analysis. You should draw across the unit content for Data Analysis—data wrangling, data visualisation, statistics, correlation and regression – to answer this assessment. Please provide your answers directly on this template. Do not delete the template wording as these assists markers to know which question you are answering. You have the option of completing this assessment either using MATLAB, Python or Excel.

Investigating Bikeway Patronage for Council Planning

Brisbane City Council (BCC) is considering upgrades to a number of bikeway networks. In order to do so, they need to understand how different bikeways are being used by both cyclists and pedestrians, so they can use this information to plan suitable bikeway upgrades. They are using data they have gathered from sensors placed along the bikeway paths, which automatically record the daily number of cyclists and pedestrians.

BCC has requested that you conduct an initial assessment regarding cyclist *and* pedestrian usage of two specific bikeway paths - Eleanor Schonell Bridge in St Lucia and Jack Pesch Bridge in Indooroopilly. In particular, BCC is investigating bikeway usage of these two bikeways paths in 2022, and if there is any relationship between different weather factors (rainfall, solar exposure and maximum temperature) and the use of these bikeway paths by both cyclists and pedestrians.

We have provided you with the BCC Bikeway Data for 2022 (between from 01 January 2022 to 31 December 2022) for daily cyclist and pedestrian usage of two locations: Eleanor Schonell Bridge in St Lucia and Jack Pesch Bridge in Indooroopilly. Furthermore, Bureau of Meteorology (BOM) for weather data captured at Brisbane City's weather station for 2022 (between from 01 January 2022 to 31 December 2022) for three weather factors: daily rainfall, solar exposure and maximum temperatures is also provided. **To assist the Python and MATLAB users with data loading the scripts to load the data in the correct format are also provided.**

Please refer to the [Assessment page in Canvas](#) for a copy of this dataset and scripts to load the data (only for Python and MATLAB users)

Remember to provide evidence to justify your responses. Evidence includes but is not limited to, any forms of visualisation, source code (for MATLAB or Python users), screenshots of Excel formulas, and screenshots of Excel "Data Analysis" tools. Please include your supporting evidence at the **end of each section** (copy and paste the code into the document, or if using live scripts merge the document PDF with the exported live script PDF). Please note your code will not be run except under exceptional circumstances, but may be used to evaluate the process that you used to answer the question.

SECTION 1 – DATA WRANGLING

1.1 Bikeway Data Inspection

- a) Create a **single** box plot visualisation to help you identify outliers across daily bikeway usage data. Your box plot should include box data for:
- Cyclist usage on Eleanor Schonell Bridge, St Lucia
 - Pedestrian usage on Eleanor Schonell Bridge, St Lucia
 - Cyclist usage on Jack Pesch Bridge, Indooroopilly
 - Pedestrian usage on Jack Pesch Bridge, Indooroopilly

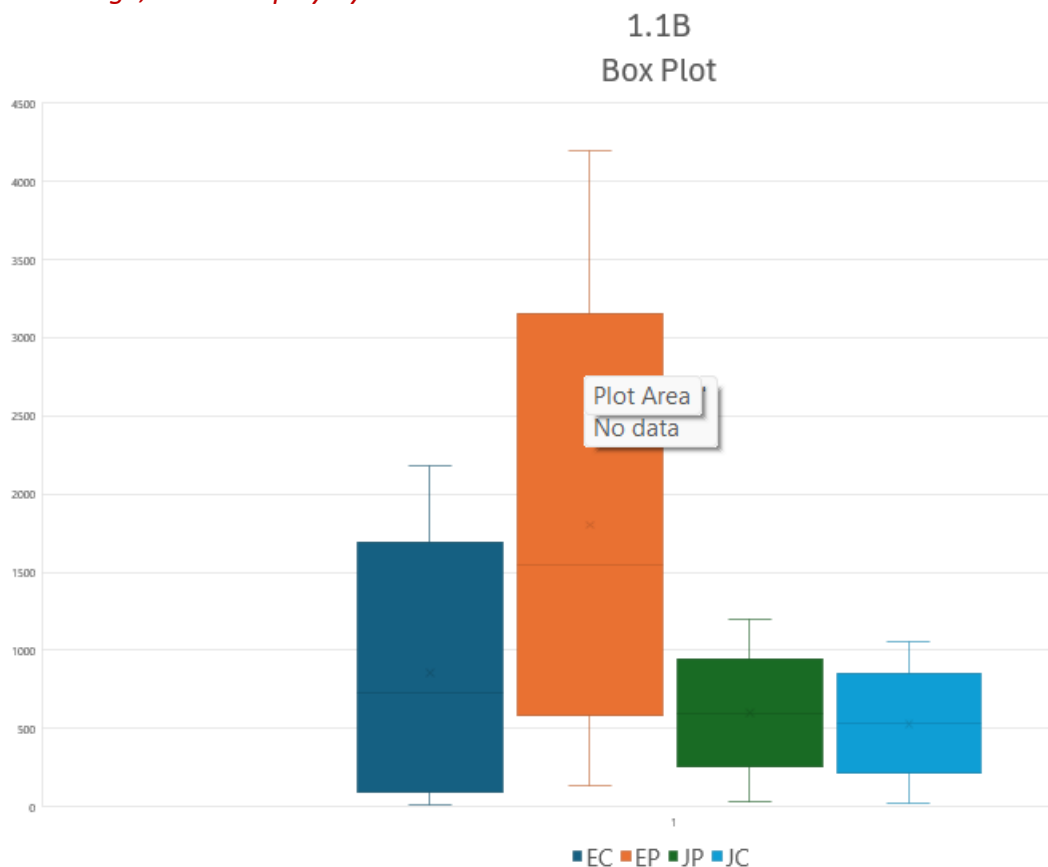
Hint: Module 8.2 - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

Eleanor Schonell Bridge, St Lucia Cyclist = EC

Eleanor Schonell Bridge, St Lucia Pedestrian = EP

Jack Pesch Bridge, Indooroopilly Pedestrian = JP

Jack Pesch Bridge, Indooroopilly Cyclist = JC

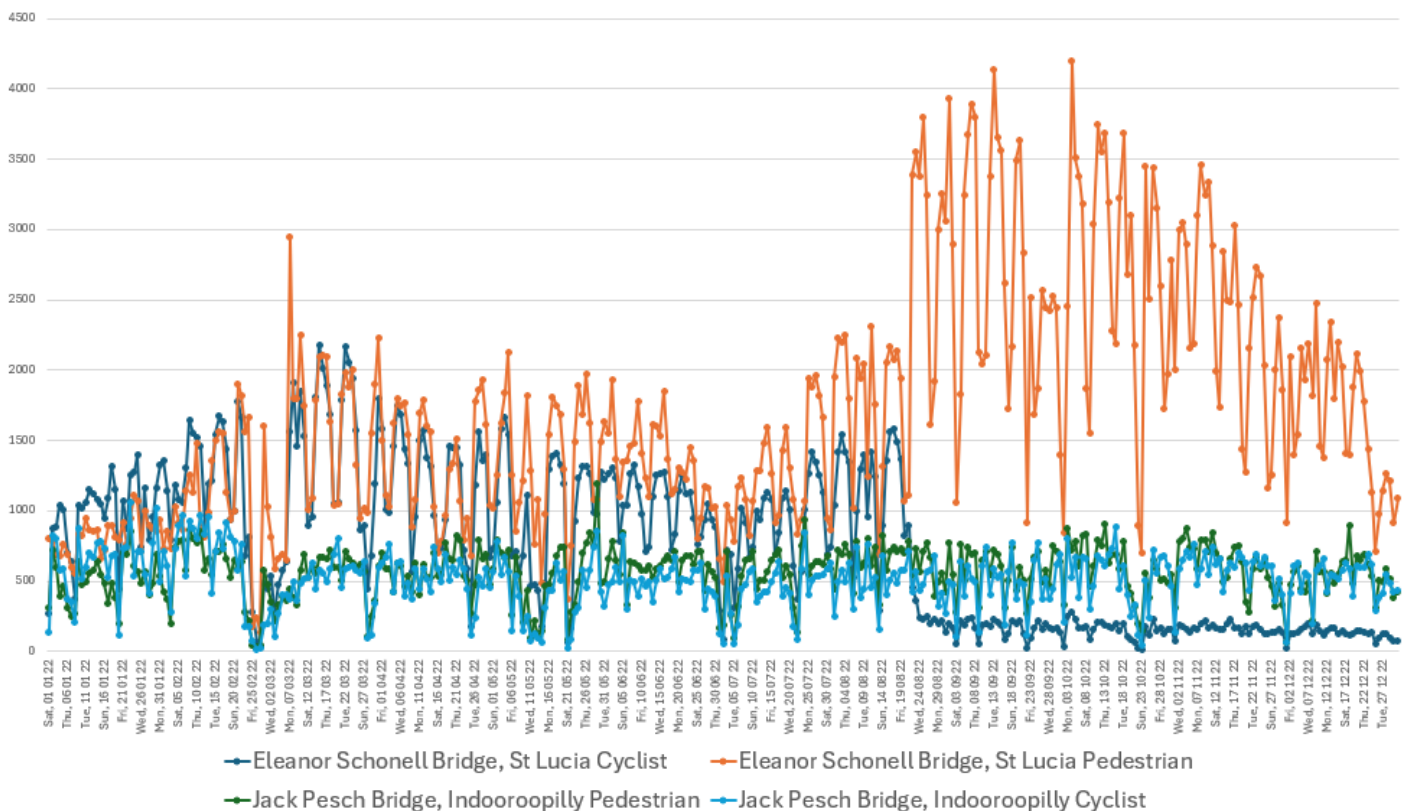


- b) Create a **single** line graph visualisation to help identify any errors, missing or other problematic data across daily bikeway usage data. Your line graph visualisation should include 4 separate lines to represent usage over time for:

- Cyclist usage on Eleanor Schonell Bridge, St Lucia
- Pedestrian usage on Eleanor Schonell Bridge, St Lucia
- Cyclist usage on Jack Pesch Bridge, Indooroopilly
- Pedestrian usage on Jack Pesch Bridge, Indooroopilly

Hint: Module 8.2 - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

1.1B
Line Graph



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- c) Based on your answers in (a) and (b) above, have you identified any problematic data (i.e. outliers, missing data, errors, etc.) in your bikeways data? If yes, please identify the problems you have found, the dates of the problematic data points, and how you intend to handle the problematic data in your final (cleaned) dataset? Provide your answers in the table below (you can add more rows if needed).

Hint: Module 8.2 - Introduction to Visualising Data & Statistics could be beneficial as you develop your response.

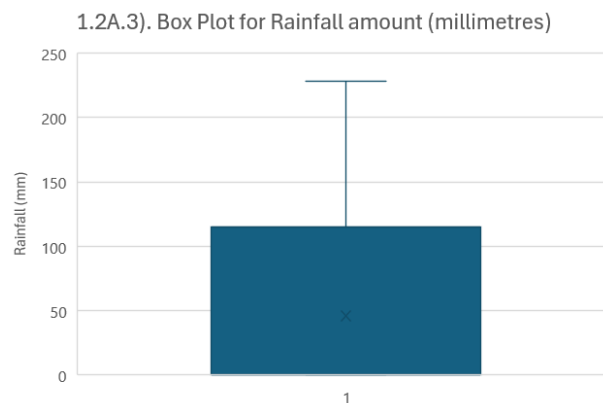
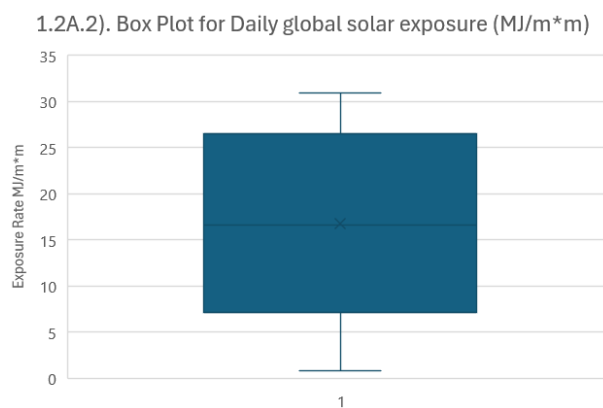
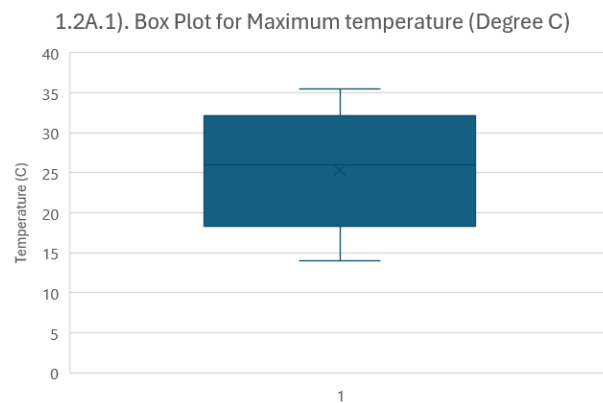
Please list all problematic data (you can add more rows if needed)	Specific date(s) and/or date range with problematic data	How do you intend to handle the problematic data identified?
Outlier for EP seems to be too big compared to the Q3	TUESDAY, 04/10/22	Lower value to 'Q3'
There is a period of time where the number for EP is double of which is seen before	Start Date: MONDAY, 22/08/22 End Date: FRIDAY, 18/11/22	Divide values above 2500, to match with rest of data.
There is a sudden high spike in EP.	MONDAY, 07/03/22	Lower value to Mean
There is a sudden drop in number of EC	Start Date: MONDAY, 22/08/22 End Date: SATURDAY, 31/12/22	Double most values.

- d) Please upload a copy of your final clean Brisbane City Council bikeways dataset (i.e. post data wrangling where all problematic data has been handled) to Canvas as a .csv or .xls file. Please use your final clean bikeways dataset for all remaining sections for this assessment.

1.2 Weather Data Inspection

- a) Create **separate boxplots** for each weather factor to help identify outliers for:
- daily rainfall in 2022
 - daily maximum temperatures in 2022
 - daily solar exposure in 2022

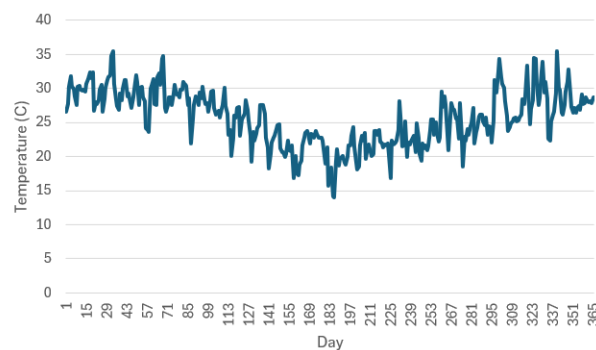
Hint: Module 8.2 - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.



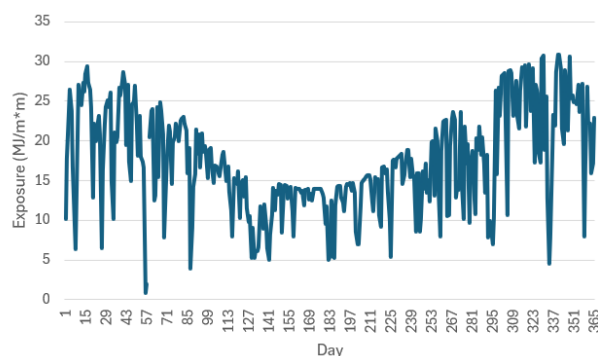
- b) Create **separate line graphs** to help identify any errors, missing or other problematic data across time for each of the following weather factors
- daily rainfall in 2022
 - daily maximum temperatures in 2022
 - daily solar exposure in 2022

Hint: Module 8.2 - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

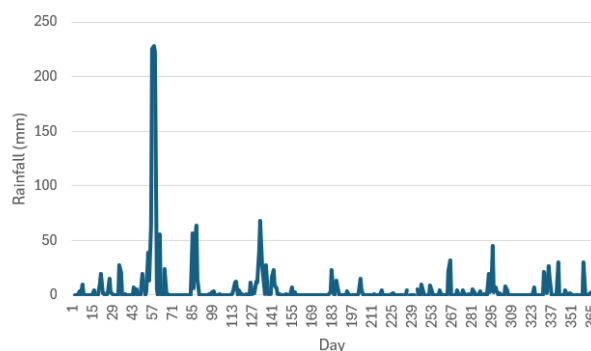
1.2B.1). Line Graph for Maximum temperature
(Degree C)



1.2B.2). Line Graph for Daily global solar exposure
(MJ/m*m)



1.2B.3). Line Graph for Rainfall amount
(millimetres)



EGH404 – Assessment 1B Template

- c) Based on your answers in (a) and (b) above, have you identified any problematic data (i.e. outliers, missing data, errors, etc.) in your weather data? If yes, please identify the problems you have found, the dates of the problematic data points, and how you intend to handle the problematic data in your final (cleaned) dataset? Provide your answers in the table below (you can add more rows if needed).

Please list all Problematic Data (you can add more rows if needed)	Specific date(s) and/or date range with problematic data	How do you intend to handle the problematic data identified?
2 blank data entries within Daily global solar exposure	Blank 1: 27/02/2022 Blank 2: 12/11/2022	Input average Exposure
5 blank data entire within Rainfall amount	Blank 1: 1/01/2022 Blank 2: 2/01/2022 Blank 3: 25/08/2022 Blank 4: 30/08/2022 Blank 5: 6/09/2022	Input average Rainfall
Very big spike in rainfall	Start: 26/02/2022 End: 28/02/2022	Divide values by 2
Lower than expected Exposure data around beginning and end of year.	Date 1: 7/01/2022 Date 2: 8/01/2022 Date 3: 26/01/2022 Date 4: 25/02/2022 Date 5: 26/02/2022 Date 6: 10/03/2022 Date 7: 19/10/2022 Date 8: 20/10/2022 Date 9: 21/10/2022 Date 10: 22/10/2022 Date 11: 30/11/2022 Date 12: 1/12/2022 Date 13: 24/12/2022	Double the values

- d) Please upload a copy of your final clean weather dataset (i.e. post data wrangling where all problematic data has been handled) to Canvas as a .csv or .xls file. Please use your final clean weather dataset for all remaining sections for this assessment.

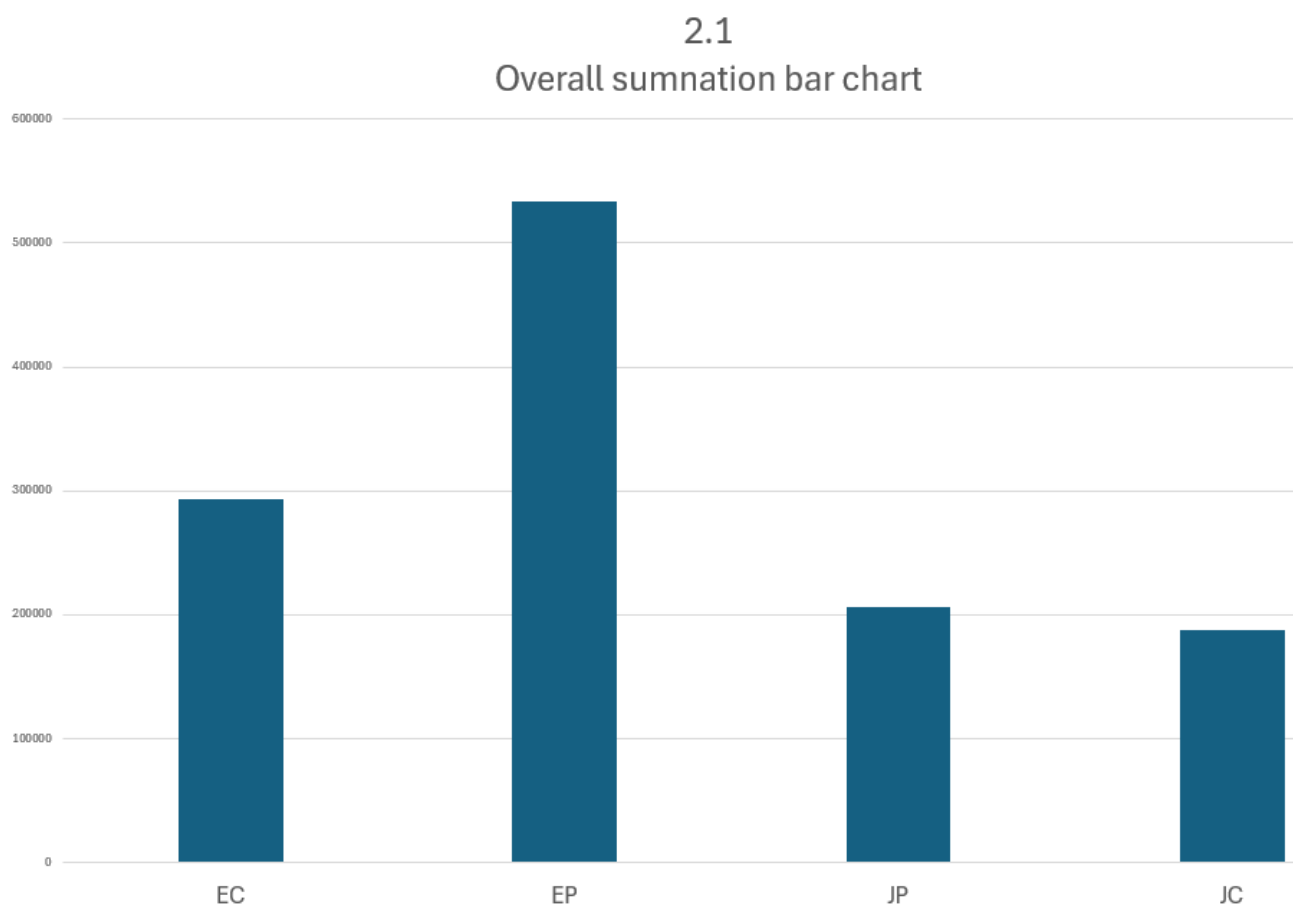
SECTION 2 – STATISTICS

Using your cleaned data, please answer the following questions.

2.1 Bikeway Usage

- a) Create a single **bar chart** visualisation that compares the yearly cyclist and pedestrian usage on Eleanor Schonell Bridge and Jack Pesch Bridge in Indooroopilly. Plot usage numbers on the y-axis and usage type (cyclist, pedestrian) on the x-axis.

Hint: Module 8.2 - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.



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b) Please answer each question below and use the table to provide your answers

Hint: Module 8.2 - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

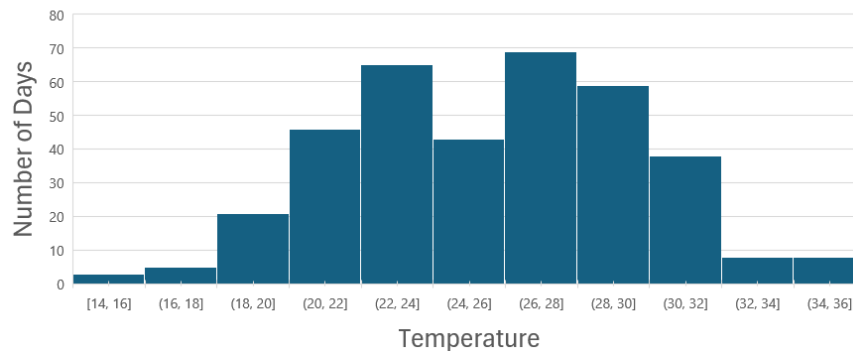
Question:	Eleanor Schonell Bridge, St Lucia	Jack Pesch Bridge, Indooroopilly
What is the most popular date for cyclists ? Please provide the number of cyclists and the date.	2181.4 (TUESDAY 15/03/22)	1058 (SUNDAY 23/01/22)
What is the most popular date for pedestrians ? Please provide the number of pedestrians and the date.	2737 (WEDNESDAY 23/11/22)	1198 (SUNDAY 29/05/22)
Using standard deviations, what month is showing the highest variability for cyclist use? Please provide the month and the standard deviation value.	606.1229 (MARCH)	315.4426 (FEBURARY)
Using standard deviations, what month is showing the lowest variability for pedestrian use? Please provide the month and the standard deviation value.	127.3419 (JANURARY)	98.79376 (JUNE)
What is the median number of cyclists?	732	537
What is the median number of pedestrians?	1487	592

2.2 Weather Factors

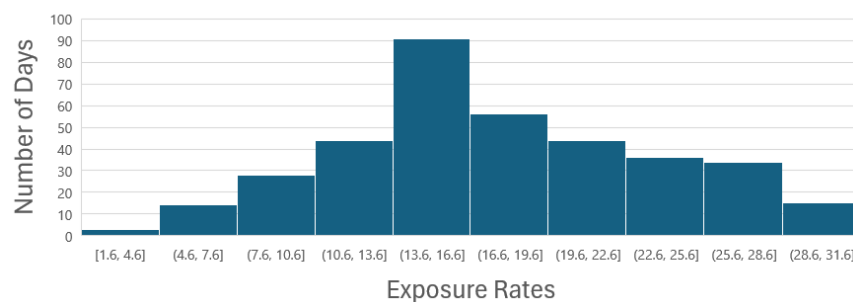
- (a) Please plot a separate **histogram** for each of your 2022 daily weather factors – rainfall, solar exposure and maximum temperature

Hint: Module 8.2 - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

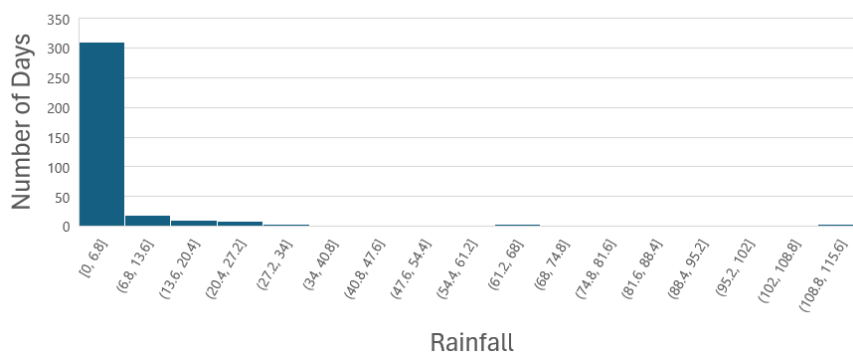
2.2A.1). Histogram for Maximum Temperature



2.2A.2). Histogram for Daily Global Solar Exposure



2.2.A.3). Histogram for Rainfall Amount



EGH404 – Assessment 1B Template

(b) Please use the table below to provide descriptive statistics for each weather factor

Hint: Module 8.2 - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

Please provide the following descriptive statistics for each weather factor using your clean weather dataset	Daily Rainfall	Solar Exposure	Maximum Temperature
Average	4.728219	17.59507	25.69041
Median	0	16.8	26
Range (i.e. minimum to maximum)	114.2	29.3	21.5
Using standard deviations, which month has the highest variability for the weather factor? Please include the month and the standard deviation value.	35.81176 (FEBURARY)	6.768309 (FEBURARY)	3.596498 (OCTOBER)
Using standard deviations, what month has the lowest variability for the weather factor? Please include the month and the standard deviation value.	1.289676 (JUNE)	1.470089 (JUNE)	1.708738 (January)

SECTION 3 – CORRELATION ANALYSIS

Using your cleaned data, please answer the following questions.

BCC wants to further investigate the relationship between daily cyclist and pedestrian usage on the Eleanor Schonell Bridge in St Lucia for 2022. Please answer the following questions:

3.1 Correlation Coefficient

Please compute the correlation coefficients for each of the relationships listed in the table below.

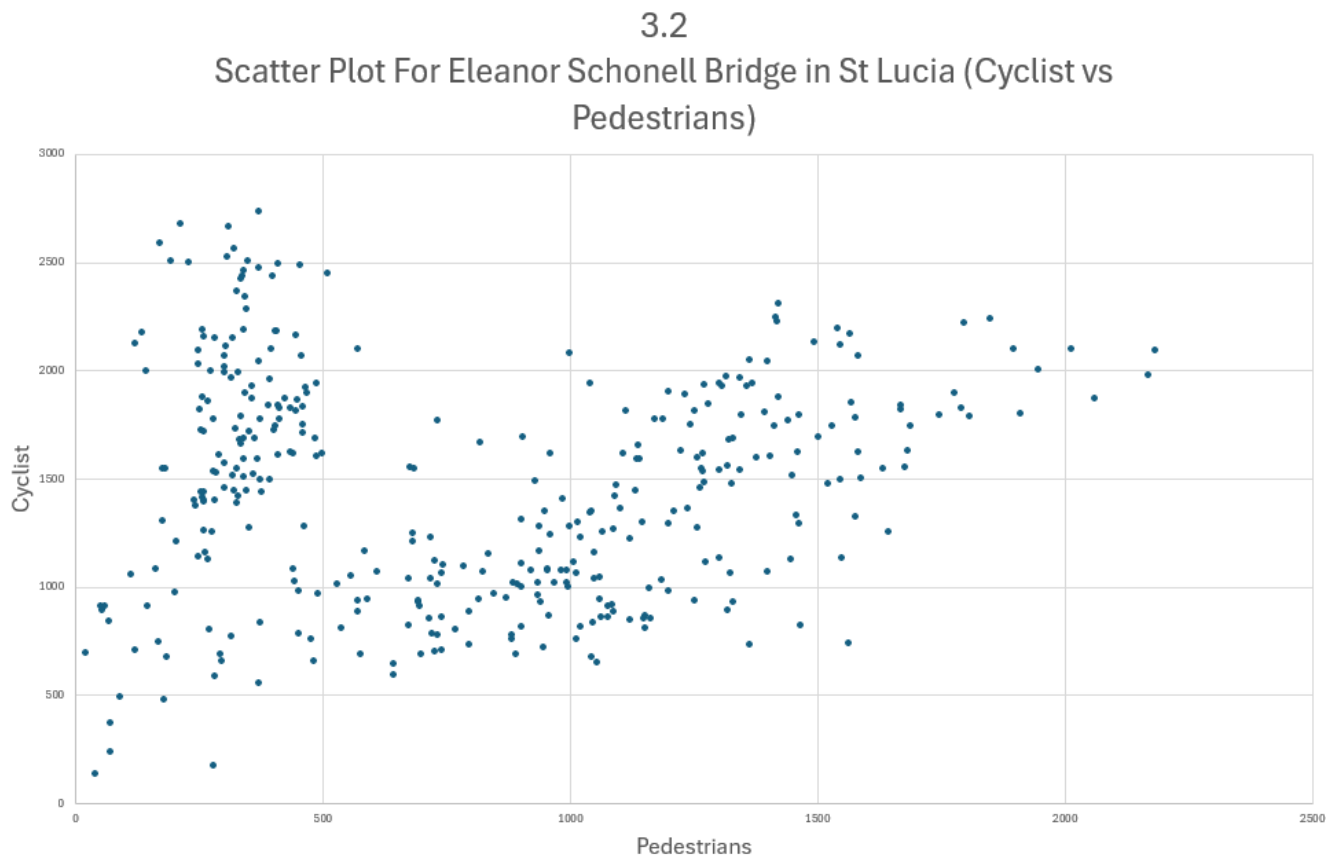
Hint: Module 9.1 - Correlation Analysis along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

Please compute the correlation coefficient between:	Correlation coefficient
(a) Cyclists and Pedestrians on the Eleanor Schonell Bridge in St Lucia	0.046707
(b) Cyclists and Pedestrians on the Jack Pesch Bridge in Indooroopilly	0.786739
(c) Cyclists on Eleanor Schonell Bridge in St Lucia and Cyclists on the Jack Pesch Bridge in Indooroopilly	0.310928
(d) Pedestrians on Eleanor Schonell Bridge in St Lucia and Pedestrians on the Jack Pesch Bridge in Indooroopilly	0.339229

3.2 Correlation Visualisation

Please create a scatter plot to visualise the relationship between daily cyclist usage (y-axis) and daily pedestrian usage (x-axis) for the Eleanor Schonell Bridge in St Lucia.

Hint: Module 9.1 - Correlation Analysis along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.



3.3 Correlation Analysis Interpretation (400 words)

Using the correlation coefficient calculated in 3.1(a) and the visualisation produced in 3.2 above, what is your interpretation of the relationship between cyclist and pedestrian usage on the Eleanor Schonell Bridge bikeway in 2022? Please use your visualisation and correlation coefficient outputs to support and interpret your interpretation of usage on the Eleanor Schonell Bridge.

Hint: Module 9.1 - Correlation Analysis could be beneficial as you develop your response.

The visual representation of the relationship between cyclist and pedestrians on the Eleanor Schonell Bridge is seen above in the scatter plot in section 3.2 and the correlation coefficient for this scatter plot is also calculated in section 3.1. The scatter plot shows some relationship between pedestrians and cyclists, but the correlation coefficient calculated says that there should be no relationship between them. However, as seen in the scatter plot, there are multiple clusters, representing some correlation between the cyclists and pedestrians. Very dense clusters represent that the correlation between them is very strong, whilst stretched-out cluster represent some correlation and other points outside clusters have no correlation.

There is one very dense cluster towards the middle left of the plot, which somewhat represents a positive correlation between cyclist and pedestrians. This cluster displays that when the number of pedestrians is below 500, the number of cyclists increases, thus representing the positive correlation between the cyclist and pedestrians. There is stretched-out cluster in the middle of the plot, which represents very little positive correlation between cyclist and pedestrians. This cluster displays that when the number of pedestrians is above 500, the number of cyclists start to spread-out and the correlation between them becomes very small or even no correlation.

With these two clusters being displayed, the reason the correlation coefficient for the overall plot is near no correlation is caused by the spread-out cluster. The dense cluster would have a very high positive correlation (maybe between 0.5 and 0.95) but this other cluster would be very low too no correlation (between 0.0005 and 0). This cluster is what would bring the overall correlation coefficient down to the cyclist and pedestrians having no correlation between them, unlike what the correlation between cyclist and pedestrians on Jack Pesch Bridge.

The correlation between cyclist and pedestrian on Jack Pesch Bridge is very high and that is seen visually in Appendix A and is calculated in section 3.1. Instead, this Scatter plot shows one major cluster, showing a clear correlation between them.

SECTION 4 – MULTIPLE LINEAR REGRESSION ANALYSIS

Using your cleaned data, please answer the following questions.

BCC wants to further investigate the relationship how daily weather factors (i.e. rainfall, solar exposure, maximum temperature) influence usage Jack Pesch Bridge, Indooroopilly – and whether there is any difference in how weather influences cyclist versus pedestrian usage of this bikeway.

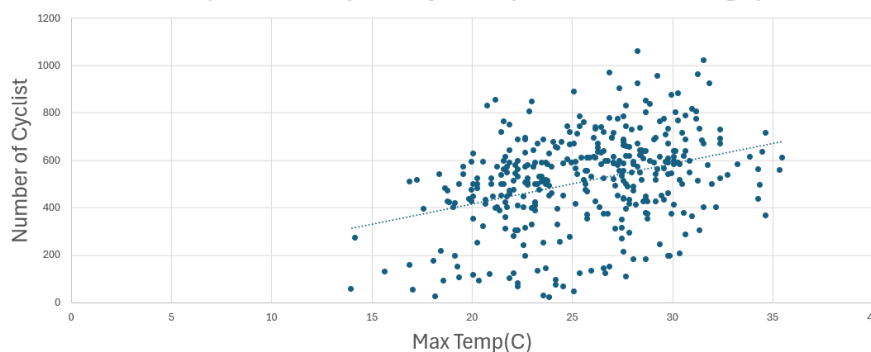
4.1 Multiple Regression Models & Supporting Visualisations

(a) Please conduct Multiple Regression analyses to investigate if any daily weather factors (i.e. rainfall, solar exposure, maximum temperature) influence **cyclist usage on the Jack Pesch Bridge, Indooroopilly**. Please include visualisations* to accompany your multiple regression models and remember to provide evidence to justify your response.

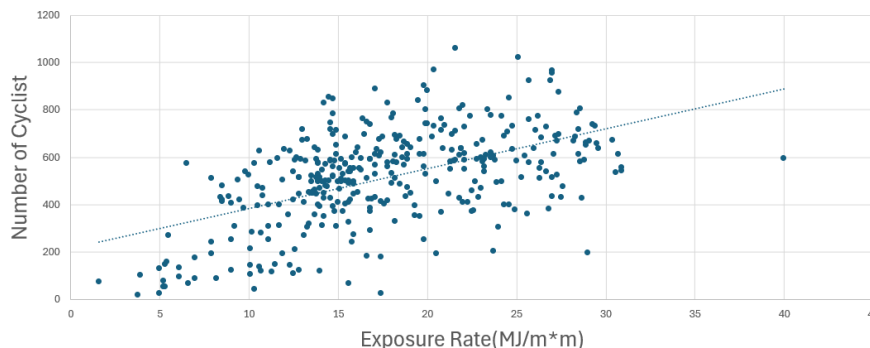
(* produce scatter plots with usage type on the y-axis and weather data factor on the x-axis and add slope line)

Hint: Module 10.1 - Regression Analysis along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

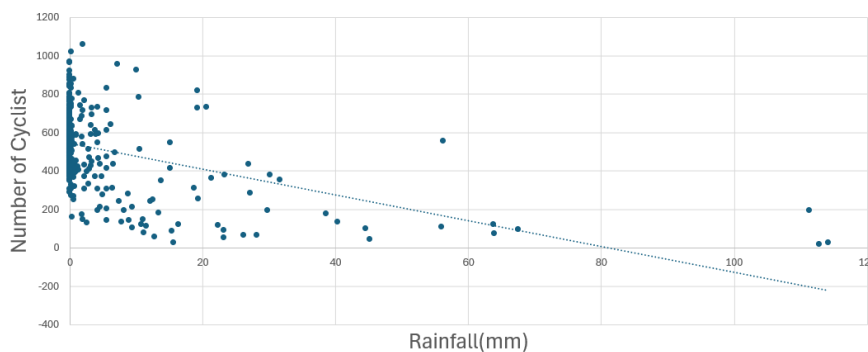
4.1A.1). MaxTemp vs Cyclist (JackPesch Bridge)



4.1A.2). Exposure vs Cyclist (JackPesch Bridge)



4.1A.3). Rainfall vs Cyclist (JackPesch Bridge)

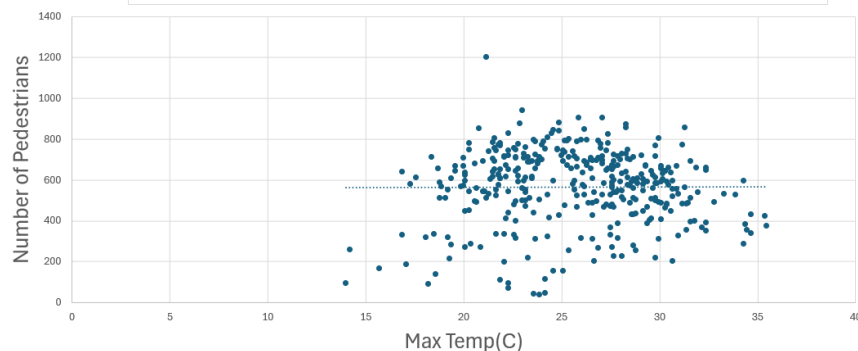


(b) Please conduct Multiple Regression analyses to investigate if any daily weather factors (i.e. rainfall, solar exposure, maximum temperature) influence **pedestrian usage on the Jack Pesch Bridge, Indooroopilly**. Please include visualisations* to accompany your multiple regression models and remember to provide evidence to justify your response.

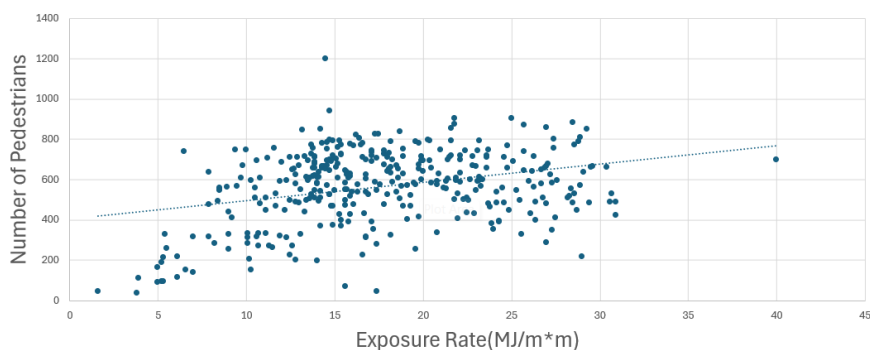
(*produce scatter plots with usage type on the y-axis and weather data factor on the x-axis and add slope line)

Hint: Module 10.1 - Regression Analysis along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

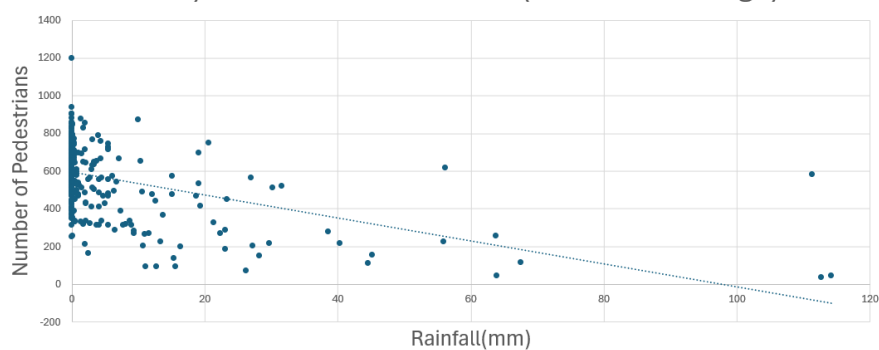
4.1B.1). MaxTemp vs Pedestrian (JackPesch Bridge)



4.1B.2). Exposure vs Pedestrian (JackPesch Bridge)



4.1B.3). Rainfall vs Pedestrian (JackPesch Bridge)



4.2 Multiple Regression Interpretation (400 words)

What is your interpretation of the results produced from the two Multiple Regression Models above, and why? Please use outputs produced from the Multiple Linear Regression models in 4.1 above to support and justify your interpretation, e.g. regression coefficients, R^2 , and p-values.

Hint: Module 10.1 - Regression Analysis could be beneficial as you develop your response.

The visual representation of the relationship between cyclist and pedestrians on the Jack Pesch Bridge against the multiple weather conditions is seen above in the scatter plots in section 4.1. The correlation and regression coefficients, as well as R^2 were also calculated. The scatter plots shows some relationship between pedestrians and cyclists and how certain weather conditions affect the usage of these bridges. As seen the correlation between them is varying depending on the type of usage and the weather condition. As seen below these are the correlation coefficients. As seen below the max temperature doesn't correlate at all with pedestrians, whilst the max temperature has a positive correlation with cyclist. The rainfall amount does correlate the same amount with pedestrians and cyclist, meaning rainfall mostly effect the usage of the bridge. The exposure does correlate with cyclist and pedestrians but are different in how much they correlate. Usage by pedestrians seem to correlate less with exposure than cyclist. These values can be seen on the slope line on the scatter plots in 4.1.

- Pedestrian:
 - Max Temp: 0.002165594
 - Exposure: 0.317150705
 - Rainfall: -0.47031088
- Cyclist:
 - Max Temp: 0.350646707
 - Exposure: 0.534037669
 - Rainfall: -0.4744625

Although the correlation shows the relationship between two variables, it doesn't show if one variable is dependent on the other variable, which is called regression. Multi-regression analysis has been calculated and visualised as seen in this section and below. As seen in the multiple scatter plots in this section the usage of the bridge by both cyclists and pedestrians is somewhat dependent on the amount of rainfall. The usage of the bridge by cyclists is more dependent on the solar exposure, unlike the pedestrians whole don't depend on the solar exposure. The usage of the bridge by cyclists is more dependent on the max temperature. However, usage of the bridge by pedestrians is less dependent on the max temperature.

- Pedestrian:
 - Max Temp:
 - $R = 0.000049$
 - $R^2 = 0.000005$
 - Exposure:
 - $R = 0.010989$
 - $R^2 = 0.100585$
 - Rainfall:
 - $R = -0.036295$
 - $R^2 = 0.221192$
- Cyclist:
 - Max Temp:
 - $R = 0.007220$
 - $R^2 = 0.122953$
 - Exposure:
 - $R = 0.016908$
 - $R^2 = 0.285196$
 - Rainfall:
 - $R = -0.033457$
 - $R^2 = 0.225115$

SECTION 5 – REFLECTION ON DATA ANALYSIS (200 words)

Please reflect on your data analysis learnings from the unit in the context of research in engineering. Briefly describe how you could use Descriptive Statistics and/or; Correlation Analysis, and/or; Regression Analysis to inform research in your engineering discipline. Please include examples of data that could be *measured to support your answer.

Hint 1- to help you answer this question assume you have collected data related to your research topic from Assessment 1 and 2.

Hint 2: Modules 7.1 – Introduction to Data analysis in Engineering, and the introductions provided in 9.1 - Correlation Analysis and 10.1 - Regression Analysis, could be beneficial as you develop your response.

This data analysis assessment could help with multiple problems within every engineering discipline. Using the research topic used for the previous two assessment this type of data analysis can be used to find discrepancy and trends in the analysis of different systems like:

- *Correlation:*
 - *How weather may affect object detection for a FOD detection system.*
 - *How many FOD at varying airports (busy or international, moderate, domestic)*
- *Regression:*
 - *Can weather predict how often a FOD incident occurs*
 - *Can FOD be based on runway traffic*

Using the engineering discipline, software engineering, there are even more data analysis that could be undertaken like:

- *Correlation:*
 - *Relation between developer experience and time to deliver*
 - *Relation between lines of code and number of bugs*
 - *Relation between software updates and security breaches*
 - *Relation between memory usage and response time.*
- *Regression*
 - *Can predict number of bugs based on complexity, size and number of contributors*
 - *Can model relationships between project size, team size and time to deliver*
 - *Can predict user satisfaction based on features, performance, testing and usability.*
 - *Can predict response time of web applications based on databases, networks and servers.*

SECTION 6 – APPENDICES

Appendix A: Scatter plot between cyclist and pedestrian on Jack Pesch Bridge

3.2Extra

Scatter Plot for Jack Pesch Bridge in Indooroopilly (Cyclist vs Pedestrians)

